



National University of Computer & Emerging Sciences, Karachi
Department of Computer Science
Final Exam Outline



Term	Phase	Sec.	Topic	Recommended Questions	CLO	Weight
Mid 1		10.1	Parametric Equations and Curves	1–34	CLO2	10%
		10.2	Calculus with Parametric Curves	1–12, 15–20, 35–40		
		10.3	Polar Coordinates and Polar Curves	1–50		
		10.4	Calculus in Polar Coordinates	1–44		
		13.2	Derivatives and Integrals of Vector Functions	1–28, 37–42		
Mid 2		14.1	Functions of Several Variables	1–36	CLO2	20%
		14.2	Limits and Continuity	1–30, 41–50	CLO1	
		14.3	Partial Derivatives	2, 9–64		
		14.4	Tangent Planes	3–10, 15–22, 31–38		
		14.5	Chain Rule	1–37	CLO1, 4	
		14.6	Directional Derivatives and Gradient Vector	1–52		
Post-Mid 2		14.7	Maxima, Minima:	5–22	CLO3, 4	70%
		14.8	Lagrange Multipliers: Gradient Descent Algorithm (Material provided)	3–16		
		15.1	Double Integrals over Rectangles	Ex: 7 — Q: 19, 22, 29, 34, 44		
		15.2	Double Integrals over General Regions	Ex: 2 — Q: 3, 9, 18, 27		
		15.6	Triple Integrals	Ex: 3, 6 — Q: 3, 7, 15, 19		
		16.2	Line Integrals	1, 3, 5, 7, 11, 15, 21, 23, 27, 35, 38		
		16.4	Green’s Theorem	5–14		
	16.5	Curl and Divergence	Ex: 1–4 — Q: 1, 3, 5, 7, 15, 17, 26, 28			